CKA-Guided Dynamic Sparse Training for Federated Learning under Non-IID Data

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Abstract

Federated learning enables decentralized model training but becomes difficult when client data are non-IID and communication is limited. Dense FedAvg communicates all parameters, while static sparse training can fail when the sparse topology is too restrictive. This project studies dynamic sparse federated learning and evaluates a CKA-guided variant, CKA-FedDST, that uses layer-wise representation similarity to adapt sparsity targets. Experiments on non-IID MNIST and Fashion-MNIST compare FedAvg, Sparse FedAvg, FedDST, and CKA-FedDST over five seeds and five sparsity levels. Dynamic sparse training is consistently more robust than fixed sparsity at high sparsity: at $s = 0.95$, FedDST reaches 0.7223 accuracy on MNIST and 0.6557 on Fashion-MNIST, while Sparse FedAvg collapses near random guessing. CKA-FedDST is stable but does not consistently improve over FedDST because measured CKA scores are highly saturated in the small-CNN setting.

1 Introduction

Federated learning (FL) trains a shared model across clients without centralizing raw data, making it attractive for privacy-sensitive and bandwidth-constrained settings [4, 10]. However, real FL clients often have heterogeneous label distributions, and this non-IID structure can destabilize local training and aggregation [5, 9]. Communication is another bottleneck: dense FedAvg sends every model parameter each round, motivating sparse and communication-efficient training [6, 13].

This project asks whether dynamic sparse training (DST) can make sparse FL more reliable under non-IID data, and whether representation similarity can guide where sparsity should be allocated. The proposed prototype, CKA-FedDST, extends FedDST [1] by periodically measuring client representation agreement with linear centered kernel alignment (CKA) [7]. Layers with higher CKA are treated as more shared and assigned lower sparsity, while lower-CKA layers are assigned higher sparsity under the same total budget. The contributions are: (1) an implementation of FedAvg, Sparse FedAvg, FedDST, and CKA-FedDST; (2) a CKA-guided layer-wise sparsity rule; and (3) a multi-seed comparison on non-IID MNIST and Fashion-MNIST.

2 Related Work

FedAvg is the canonical FL baseline, combining local client SGD with server-side model averaging [10]. Subsequent work studies communication-efficient FL and optimization under client drift, including federated optimization methods [6], local SGD analysis [13], FedProx [9], and SCAFFOLD [5]. These works motivate both the baseline choice and the non-IID setting used here.

Sparse neural networks reduce parameter and communication costs. The Lottery Ticket Hypothesis showed that trainable sparse subnetworks can exist inside dense models [3]. Sparse evolutionary training and RigL use prune-and-regrow dynamics to maintain sparse networks during training [2, 11]. FedDST adapts this idea to FL by reducing both client computation and communication [1]. CKA and related representation-similarity tools provide a way to compare internal activations across models [7, 12]. This project combines these two directions by using CKA as a heuristic signal for layer-wise sparse allocation. MNIST and Fashion-MNIST provide controlled grayscale benchmarks for this prototype [8, 14].

Table 1: Final accuracy at dense or extreme sparsity. Mean ± standard deviation over five seeds.

Dataset	Method	Sparsity	Final accuracy
MNIST	FedAvg	dense	0.9415 ± 0.0167
MNIST	Sparse FedAvg	0.95	0.1064 ± 0.0065
MNIST	FedDST	0.95	0.7223 ± 0.0663
MNIST	CKA-FedDST	0.95	0.6908 ± 0.1547
Fashion	FedAvg	dense	0.7795 ± 0.0172
Fashion	Sparse FedAvg	0.95	0.1000 ± 0.0000
Fashion	FedDST	0.95	0.6557 ± 0.0264
Fashion	CKA-FedDST	0.95	0.6407 ± 0.0470

3 Experiments

Setup. Experiments use MNIST and Fashion-MNIST, each split across five clients with Dirichlet label-skew partitioning at $\alpha = 0.3$. The model is a small CNN with two convolutional layers and two fully connected layers. Training uses 20 communication rounds, one local epoch, batch size 64, learning rate 0.01, and seeds $\{7, 13, 21, 42, 100\}$. Sparse methods use unstructured weight masks at sparsity $s \in \{0.5, 0.7, 0.8, 0.9, 0.95\}$; biases are not sparsified. FedDST and CKA-FedDST update masks every two rounds by pruning low-magnitude active weights and regrowing inactive weights with high gradient magnitude. CKA-FedDST computes CKA on a balanced 200-example reference set using activations from `conv1`, `conv2`, and `fc1`. The multi-seed suite contains 80 runs per dataset, and the CKA-strength suite contains 125 CKA-FedDST runs per dataset over strengths $\{0.2, 0.5, 0.8, 0.9, 1.0\}$. All sparse methods preserve the requested total sparsity; $s = 0.95$ leaves about 21k active maskable weights.

Results. Figure 1 and Table 1 show the central trend. Fixed Sparse FedAvg works at moderate sparsity but fails at $s = 0.95$, reaching only 0.1064 on MNIST and 0.1000 on Fashion-MNIST. FedDST preserves useful performance at the same sparsity, improving over Sparse FedAvg by +0.6159 on MNIST and +0.5557 on Fashion-MNIST. CKA-FedDST follows FedDST closely but is not consistently better: it helps slightly at some intermediate Fashion-MNIST sparsities, but underperforms FedDST at $s = 0.95$. CKA-strength sweeps from 0.2 to 1.0 change average accuracy only marginally.

4 Discussion

The experiments support a clear conclusion: dynamic sparse training is much more robust than a fixed sparse topology in non-IID FL, especially under extreme sparsity. FedDST's prune-and-regrow

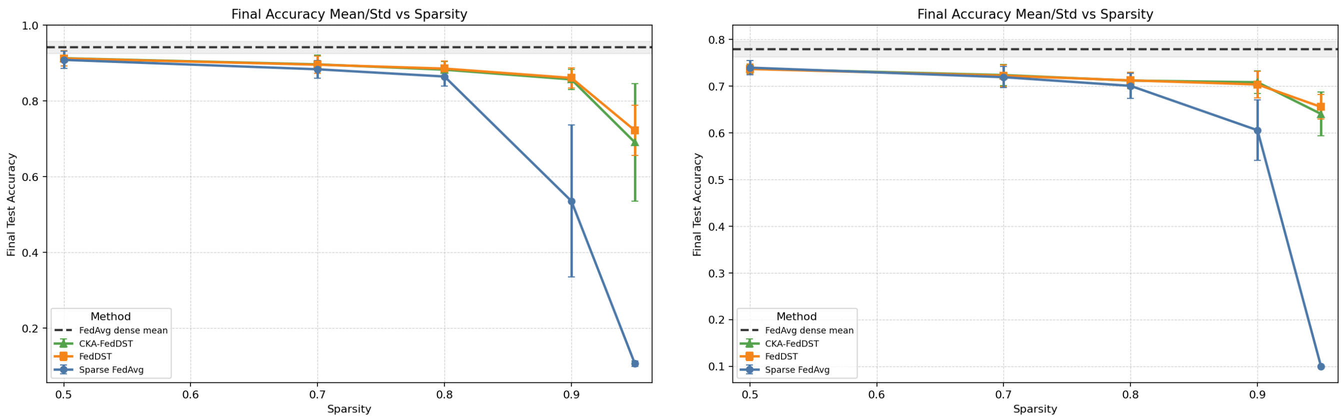

Figure 1: Final test accuracy versus sparsity on MNIST (left) and Fashion-MNIST (right). Dynamic sparse methods remain trainable at high sparsity.

mechanism can recover useful subnetworks during training, while Sparse FedAvg is constrained by its initial mask.

The CKA-guided idea is scientifically plausible but not yet empirically strong in this prototype. Layer-wise CKA values are saturated: `conv1` and `conv2` are almost always near 1.0, and `fc1` remains high on both datasets. With little contrast across layers, CKA has limited ability to change sparsity targets meaningfully. Future work should test deeper models and harder datasets, such as CIFAR-10 with a ResNet, more clients, stronger non-IID settings, explicit communication-cost measurement in bits, and alternative signals such as gradient similarity or representational similarity analysis. Thus, the project demonstrates a working CKA-FedDST prototype, but the strongest supported result is the value of dynamic sparsity rather than CKA guidance itself.

References

[1] Sameer Bibikar, Haris Vikalo, Zhangyang Wang, and Xiaohan Chen. 2022. Federated Dynamic Sparse Training: Computing Less, Communicating Less, Yet Learning Better. *Proceedings of the AAAI Conference on Artificial Intelligence* 36, 6 (2022), 6080–6088. doi:10.1609/aaai.v36i6.20555

[2] Utku Evci, Trevor Gale, Jacob Menick, Pablo Samuel Castro, and Erich Elsen. 2020. Rigging the Lottery: Making All Tickets Winners. In *Proceedings of the 37th International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 119)*. PMLR, 2943–2952.

[3] Jonathan Frankle and Michael Carbin. 2019. The Lottery Ticket Hypothesis: Finding Sparse, Trainable Neural Networks. In *International Conference on Learning Representations*.

[4] Peter Kairouz, H. Brendan McMahan, Brendan Avent, Aurélien Bellet, Mehdi Bennis, Arjun Nitin Bhagoji, Kallista Bonawitz, Zachary Charles, Graham Cormode, Rachel Cummings, et al. 2021. Advances and Open Problems in Federated Learning. *Foundations and Trends in Machine Learning* 14, 1–2 (2021), 1–210.

[5] Sai Praneeth Karimireddy, Satyen Kale, Mehryar Mohri, Sashank Reddi, Sebastian U. Stich, and Ananda Theertha Suresh. 2020. SCAFFOLD: Stochastic Controlled Averaging for Federated Learning. In *Proceedings of the 37th International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 119)*. PMLR, 5132–5143.

[6] Jakub Konečný, H. Brendan McMahan, Daniel Ramage, and Peter Richtárik. 2016. Federated Optimization: Distributed Machine Learning for On-Device Intelligence. *arXiv preprint arXiv:1610.02527* (2016).

[7] Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. 2019. Similarity of Neural Network Representations Revisited. In *Proceedings of the 36th International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 97)*. PMLR, 3519–3529.

[8] Yann LeCun, Léon Bottou, Yoshua Bengio, and Patrick Haffner. 1998. Gradient-Based Learning Applied to Document Recognition. *Proc. IEEE* 86, 11 (1998), 2278–2324.

[9] Tian Li, Anit Kumar Sahu, Manzil Zaheer, Maziar Sanjabi, Ameet Talwalkar, and Virginia Smith. 2020. Federated Optimization in Heterogeneous Networks. In *Proceedings of Machine Learning and Systems*, Vol. 2. 429–450.

[10] H. Brendan McMahan, Eider Moore, Daniel Ramage, Seth Hampson, and Blaise Agüera y Arcas. 2017. Communication-Efficient Learning of Deep Networks from Decentralized Data. In *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics (Proceedings of Machine Learning Research, Vol. 54)*. PMLR, 1273–1282.

[11] Decebal Constantin Mocanu, Elena Mocanu, Peter Stone, Phuong H. Nguyen, Madeleine Gibescu, and Antonio Liotta. 2018. Scalable Training of Artificial Neural Networks with Adaptive Sparse Connectivity Inspired by Network Science. *Nature Communications* 9, 1 (2018), 2383.

[12] Maithra Raghu, Justin Gilmer, Jason Yosinski, and Jascha Sohl-Dickstein. 2017. SVCCA: Singular Vector Canonical Correlation Analysis for Deep Learning Dynamics and Interpretability. In *Advances in Neural Information Processing Systems*, Vol. 30.

[13] Sebastian U. Stich. 2019. Local SGD Converges Fast and Communicates Little. In *International Conference on Learning Representations*.

[14] Han Xiao, Kashif Rasul, and Roland Vollgraf. 2017. Fashion-MNIST: A Novel Image Dataset for Benchmarking Machine Learning Algorithms. *arXiv preprint arXiv:1708.07747* (2017).